

SECTION 5.5

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d) The matrices $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$ are not similar because $\text{tr} A \neq \text{tr} B$: $\text{tr} A = 5$ and $\text{tr} B = 4$.

e) The matrices $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -2 & 1 \\ -2 & 4 & -2 \\ -3 & 6 & -3 \end{bmatrix}$ are not similar because $C_A \neq C_B$:

$$C_A(x) = \begin{vmatrix} x-2 & -1 & -1 \\ -1 & x & -1 \\ -1 & -1 & x \end{vmatrix} = x^3 - 2x^2 - 3x, \quad C_B(x) = \begin{vmatrix} x-1 & 2 & -1 \\ 2 & x-4 & 2 \\ 3 & -6 & x+3 \end{vmatrix} = x^3 - 2x^2$$

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c) $A = \begin{bmatrix} 3 & 1 & 6 \\ 2 & 1 & 0 \\ -1 & 0 & -3 \end{bmatrix}$

• Eigenvalues of A : solutions to $C_A(x) = 0$

$$C_A(x) = \begin{vmatrix} x-3 & -1 & -6 \\ -2 & x-1 & 0 \\ 1 & 0 & x+3 \end{vmatrix} = x^3 - x^2 - 5x - 3 = (x+1)^2(x-3)$$

There are two eigenvalues: $\lambda_1 = -1$ (with multiplicity 2) and $\lambda_2 = 3$ (with multiplicity 1).

• Basis of $E_{-1}(A) = \text{null}(-I - A)$:

$$-I - A = \begin{bmatrix} -4 & -1 & -6 \\ -2 & -2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \Rightarrow \text{The elements of } \text{null}(A) \text{ are the solutions to } \begin{bmatrix} -4 & -1 & -6 \\ -2 & -2 & 0 \\ 1 & 0 & 2 \end{bmatrix} \vec{x} = \vec{0}.$$

The above system has a unique basic solution $\begin{bmatrix} -2 \\ 2 \\ 1 \end{bmatrix}$, so a basis of $E_{-1}(A)$ is $\{(-2, 2, 1)\}$.

• $\dim E_{-1}(A) < \text{multiplicity of } -1 \Rightarrow A \text{ is not diagonalizable}$

" 1
" 2

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$P^{-1}\vec{x}$ is an eigenvector of B with eigenvalue $\lambda \in \mathbb{C}$: $BP^{-1}\vec{x} = \lambda P^{-1}\vec{x}$ and $P^{-1}\vec{x} \neq \vec{0}$.

Note that $\vec{x} \neq \vec{0}$ and P is invertible, so $P^{-1}\vec{x} \neq \vec{0}$. In addition, $BP^{-1}\vec{x} = P^{-1}APP^{-1}\vec{x} = P^{-1}A\vec{x} = P^{-1}(\lambda\vec{x}) = \lambda P^{-1}\vec{x} \checkmark$
 $\uparrow$
 $A\vec{x} = \lambda\vec{x}$

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Since A is diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_n$, we have $A \sim \text{diag}(\lambda_1, \dots, \lambda_n)$. This implies:

$$a) \det A = \begin{vmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{vmatrix} = \lambda_1 \lambda_2 \dots \lambda_n;$$

$$b) \text{tr } A = \text{tr} \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix} = \lambda_1 + \lambda_2 + \dots + \lambda_n$$

SECTION 6.1

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c) $V = \{p(x) \in P \mid p(x) \text{ has degree } \leq 3\} = P_3$, so it is a vector space.

d) $V = \{1, x, x^2, \dots\}$ is not a vector space because $1, x \in V$ but $1+x \notin V$.

h) $V = \mathbb{R}$ is a vector space (since $\mathbb{R} = \mathbb{R}^1$ and we know $\mathbb{R}^n$ is a vector space).

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b)

$$a\vec{u} + b\vec{v} + c\vec{w} = \vec{0} \Rightarrow a \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + b \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + c \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} a+c & b+c \\ b+c & a-c \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow \begin{cases} a+c=0 \\ b+c=0 \\ a-c=0 \end{cases} \Rightarrow a=b=c=0$$

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Gaussian elimination

c)

$$a\vec{u} + b\vec{v} + c\vec{w} = \vec{0} \Rightarrow a(x^3+x) + b(x^2+1) + c(x^3-x^2+x+1) = (a+c)x^3 + (b-c)x^2 + (a+c)x + (b+c) = \vec{0}$$

$$\Rightarrow \begin{cases} a+c=0 \\ b-c=0 \\ b+c=0 \end{cases} \Rightarrow a=b=c=0$$

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Gaussian elimination

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To prove that $-\vec{0} = \vec{0}$, we have to show that $\vec{0} + \vec{0} = \vec{0}$ (definition of the negative of a vector). We have

$$\vec{0} + \vec{0} = \vec{0} \Rightarrow -\vec{0} = \vec{0}.$$

$$\vec{0} + \vec{x} = \vec{0} \quad \text{for all } \vec{x} \in V$$

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Suppose V has two zero vectors $\vec{0}, \vec{0}' \in V$. We have to show $\vec{0} = \vec{0}'$.

By definition of zero vector, we have $\vec{0} + \vec{x} = \vec{x} + \vec{0} = \vec{0}' + \vec{x} = \vec{x} + \vec{0}' = \vec{x}$ for all $\vec{x} \in V$. Therefore:

$$\vec{0} = \vec{0} + \vec{0}' = \vec{0}'$$

$$\begin{array}{cc} \vec{0} + \vec{x} = \vec{x} & \vec{x} + \vec{0}' = \vec{x} \\ \uparrow & \uparrow \\ \text{for all } \vec{x} & \text{for all } \vec{x}. \end{array}$$

We conclude that the zero vector is unique.

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Suppose $\vec{v} \in V$ and $\vec{w}_1, \vec{w}_2 \in V$ are negatives of $\vec{v}$. This means that

$$\vec{v} + \vec{w}_1 = \vec{w}_1 + \vec{v} = \vec{v} + \vec{w}_2 = \vec{w}_2 + \vec{v} = \vec{0}$$

Thus,

$$\vec{w}_1 = \vec{w}_1 + \vec{0} = \vec{w}_1 + (\vec{v} + \vec{w}_2) = (\vec{w}_1 + \vec{v}) + \vec{w}_2 = \vec{0} + \vec{w}_2 = \vec{w}_2.$$

$$\begin{array}{ccccccc} \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow & \uparrow \\ \vec{x} = \vec{x} + \vec{0} & \vec{v} + \vec{w}_2 = \vec{0} & \text{addition} & \vec{w}_1 + \vec{v} = \vec{0} & \vec{0} + \vec{x} = \vec{x} & & \\ \text{for all } \vec{x} \in V & & \text{is associative} & & \text{for all } \vec{x} \in V. & & \end{array}$$

Therefore, $\vec{w}_1 = \vec{w}_2$, so the negative of $\vec{v}$ is unique.